VISUAL JIGSAW POST-TRAINING IMPROVES MLLMS

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ABSTRACT

Reinforcement learning (RL) post-training has recently unlocked major gains in multimodal large language models (MLLMs), yet most recipes remain text-centric: dense visual inputs are used only to support chain-of-thought, leaving intrinsic perception under-optimized. We introduce Visual Jigsaw, a simple, verifiable, and self-supervised post-training task that strengthens vision-centric understanding without any generative visual components. The key idea is to partition a visual input (image, video, or RGB-D) into elements, shuffle them, and ask the model to predict the original ordering using text-only outputs. This yields a binary verifiable objective suitable for RL with lightweight format constraints and a graded reward that combines exact-match accuracy with discounted partial correctness (gamma 0.2), which prevents reward hacking and stabilizes optimization on harder jigsaws. Instantiations across images (3x3 patches on COCO), videos (temporal clip ordering on LLaVA-Video), and 3D (depth/geometry ordering on ScanNet) consistently improve perception while preserving reasoning. On images, we observe MMVP 60.66 vs 54.66 (+6.00), MMStar (fine-grained) 65.81 vs 59.75 (+6.06), HR-Bench 71.13 vs 67.38 (+3.75), and DA-2K 60.35 vs 54.45 (+5.90). On videos (64 frames), we obtain AoTBench 57.64 vs 52.41 (+5.23), Vinoground 25.20 vs 21.80 (+3.40), VideoMME 64.74 vs 63.44 (+1.30), and CVBench 54.50 vs 51.50 (+3.00). On 3D benchmarks, we observe DA-2K 71.56 vs 54.45 (+17.11), SAT-Real 64.00 vs 48.66 (+15.34), and OmniSpatial 45.99 vs 42.66 (+3.33). Compared to supervised fine-tuning (SFT), RL yields larger and more consistent gains, with SFT sometimes degrading perception (e.g., LISA-Grounding 66.59 with SFT vs 74.54 with RL). Visual Jigsaw is plug-and-play for text-output MLLMs (e.g., Qwen2.5-VL-7B) and complements reasoning-oriented RL without harming language or math skills.

1 INTRODUCTION

Reinforcement learning (RL) post-training has accelerated progress in large language and multimodal models, but most recipes remain text-centric: images and videos are primarily used to extract sparse cues to support chain-of-thought, rather than to directly cultivate perceptual competence. As a consequence, intrinsic visual skills such as fine-grained recognition, temporal ordering, and 3D spatial reasoning often lag behind models' language reasoning ability. Recent multimodal systems like Qwen2-VL and Qwen2.5-VL have improved visual tokenization and multi-frame processing (Wang et al., 2024b; Bai et al., 2025), and RL-based reasoning lines like LLaVA-Critic-R1, VL-Cogito, and Video-R1 (Wang et al., 2025a; Yuan et al., 2025; Feng et al., 2025) show the promise of verifiable, rule-based rewards. However, a simple, general-purpose, vision-centric post-training objective that requires only text outputs and scales across modalities remains under-explored.

We propose **Visual Jigsaw**, a generic self-supervised post-training task that recasts visual understanding as permutation recovery. Given an image, video, or 3D input, we partition it into elements (e.g., patches, clips, or depth-ordered point sets), shuffle them, and train the model to predict the original order as a textual permutation. The objective is verifiable from the input alone, requires no captions, and avoids any visual generative modules. We use RL with a graded reward that mixes exact-match success with discounted partial correctness and strict format checks, which improves stability and prevents reward hacking on harder jigsaws. We instantiate Visual Jigsaw across images (COCO (Lin et al., 2014)), videos (LLaVA-Video), and 3D (ScanNet (Dai et al., 2017)).

Contributions. (1) We introduce a verifiable, self-supervised, vision-centric RL post-training task requiring only text outputs. (2) We provide general instantiations across images, videos, and 3D,

improving fine-grained, temporal, and spatial understanding. (3) We design a graded reward (partial correctness plus format constraints) that stabilizes optimization on harder jigsaws. (4) We demonstrate superior transfer to perception benchmarks versus reasoning-centric RL, while preserving reasoning ability on math and multimodal tasks (Hao et al., 2025; Yue et al., 2024; Lu et al., 2023; Wang et al., 2024a; Zhang et al., 2024b). Visual Jigsaw is plug-and-play for Qwen2.5-VL-7B-based models (Bai et al., 2025) and complements reasoning methods such as LLaVA-Critic-R1 (Wang et al., 2025a) and VL-Cogito (Yuan et al., 2025).

2 Related Works

RL-based post-training for MLLMs. Recent work adapts reinforcement learning and preference optimization to multimodal settings. Lines such as LLaVA-Critic-R1 (Wang et al., 2025a), VL-Cogito (Yuan et al., 2025), and Video-R1 (Feng et al., 2025) explore verifiable or curriculum-based objectives to improve multimodal reasoning. Direct Preference Optimization (DPO) (Rafailov et al., 2023) provides an alternative to RLHF in language models, and MLLM variants increasingly leverage rule-based rewards due to their stability and verifiability. Our work differs by focusing on a *vision-centric*, *self-supervised* objective that requires only text outputs and scales naturally from images to videos and 3D.

Base MLLMs and visual backbones. Advances in visual tokenization and multi-frame processing, exemplified by Qwen2-VL and Qwen2.5-VL (Wang et al., 2024b; Bai et al., 2025), improve perceptual fidelity and enable efficient processing of images and videos. Visual Jigsaw is orthogonal and can be applied to these models without architectural changes.

Self-supervised vision pretexts. Classic pretexts learn invariances via jigsaw, rotation prediction, and temporal order verification. Closest are jigsaw-style objectives over spatial or spatio-temporal tiles, including video variants and recent rule-based visual RL that uses jigsaw puzzles as a testbed (Wang et al., 2025c). Our use differs by serving as a verifiable post-training signal for instruction-following MLLMs with text-only outputs and rule-checked rewards, yielding cross-modal transfer without extra generators.

Perception benchmarks. We evaluate on a broad suite of perception and spatial reasoning benchmarks, including EMMA (Hao et al., 2025), MMMU (Yue et al., 2024), MathVista (Lu et al., 2023), MathVision (Wang et al., 2024a), MathVerse (Zhang et al., 2024b), Winoground (Thrush et al., 2022), SugarCrepe++ (Dumpala et al., 2024), CVBench (Zhu et al., 2025), VideoMME (Fu et al., 2025), TempCompass (Liu et al., 2024), Vinoground (Zhang et al., 2024a), AoTBench (Xue et al., 2025), OmniSpatial (Jia et al., 2025), 3DSRBench, ViewSpatial-Bench, All-Angles Bench, and DA-2K (Yang et al., 2024). Where possible, we report results with the official evaluation protocols. Our image instantiation uses COCO (Lin et al., 2014) and our 3D instantiation uses ScanNet (Dai et al., 2017) to build jigsaw tasks.

3 Method

Problem formulation. Visual Jigsaw reformulates visual understanding as recovering the correct ordering of shuffled elements derived from a visual input. For an image, we partition it into a grid of patches; for a video, into temporally contiguous clips; for RGB-D or multi-view inputs, into depth- or view-ordered subsets. The model receives the shuffled elements alongside an instruction and must output a valid permutation (e.g., a list of indices) that reconstructs the canonical order.

Task construction. We construct three modality-specific jigsaws: - Image Jigsaw: 3×3 grid over images drawn from COCO. Elements are patches; the target is the raster-scan order. - Video Jigsaw: temporal segmentation of videos (e.g., 16/32/64 frames) into equal-length clips. Elements are clips; the target is chronological order. - 3D Jigsaw: from RGB-D or multi-view frames (e.g., ScanNet), we derive depth- or geometry-aware groupings (e.g., near-to-far bins, structured point subsets). Elements are bins or subsets; the target is the physically consistent order.

Policy and output format. Parsing and validation. We canonicalize permutations with a single delimiter and no spaces; responses are validated by a small regex and mapped back to index sets before scoring. Ties or duplicates are treated as invalid to keep rewards verifiable. We use a standard instruction-following MLLM (Qwen2.5-VL-7B-Instruct (Bai et al., 2025)) and constrain outputs to

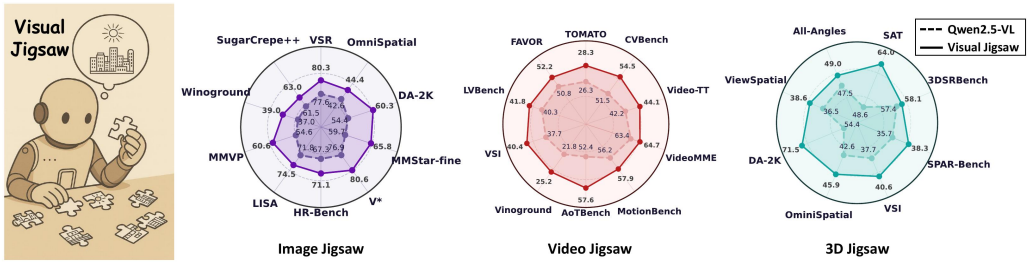

Figure 1: Visual Jigsaw overview across image, video, and 3D. Self-supervised permutation recovery with verifiable rewards.

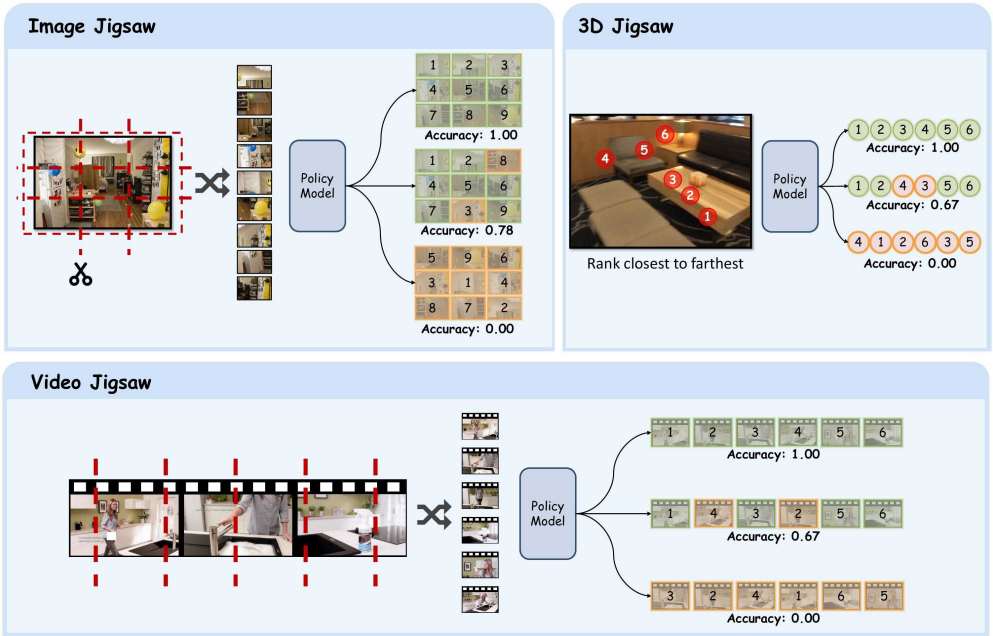

Figure 2: Illustration of the Visual Jigsaw tasks. The model predicts the original ordering of shuffled visual elements using text-only outputs.

a strict format (e.g., `[1,2,...,K]`). A lightweight parser rejects malformed outputs with zero reward, which discourages reward hacking and encourages format adherence.

Reward design. We adopt a verifiable reward with two components: (1) exact-match success when the predicted permutation equals the ground truth, and (2) partial correctness measured by normalized discounted agreement with a discount factor gamma 0.2 that down-weights later positions. The final reward is the sum of the components subject to format validity; invalid permutations receive zero reward. This graded shaping drives learning on hard jigsaws while maintaining a clear optimum at exact match.

Optimization. Reproducibility and hygiene. We fix seeds, log preprocessing decisions, and compute ground-truth order directly from the inputs; we strip filenames and EXIF metadata and randomize index conventions to avoid shortcut leakage. We optimize with group-based policy gradients (GRPO-style updates) over small batches of candidate completions per prompt, normalizing within-group returns to reduce variance. We use a short warm-up with format-only rewards, then enable the full graded reward. No visual generative modules are used, and no human or synthetic annotations are required beyond the original data sources.

Shortcut mitigation. We randomize crop offsets, remove borders, and add light photometric/geometry jitter for images; sample variable strides with temporal jitter for videos; and derive 3D bins from surface cues (normals, plane distance) to reduce sensor noise. Index conventions (row- vs column-major) are randomized per prompt to avoid pattern memorization.

4 EXPERIMENTS

Setup. Unless noted, we start from Qwen2.5-VL-7B-Instruct (Bai et al., 2025). For Image Jigsaw, we derive 3×3 patch permutations from COCO (Lin et al., 2014). For Video Jigsaw, we use segments from LLaVA-Video. For 3D Jigsaw, we construct depth-ordered groupings from ScanNet (Dai et al., 2017). We compare RL-based Visual Jigsaw to the base model and, when applicable, to an SFT variant trained to predict permutations with cross-entropy but without RL.

Training details. We mix image, video, and 3D jigsaw prompts within each epoch with a 4:3:1 ratio, sample $K \in \{4, 6, 9\}$ for images and $\{4, 8\}$ clips for videos, and randomize near/far bin counts for 3D. We generate 4 candidates per prompt, compute verifiable rewards, and apply GRPO updates with within-group normalization. A short curriculum increases difficulty over epochs by expanding K and clip counts while keeping the same prompt format.

Evaluation protocol. We use official evaluators when available and fix prompts, token budgets, and frame samplers across baselines. SFT uses the same data and format but optimizes cross-entropy alone. We report average accuracy over 3 seeds when evaluators are stochastic and include bootstrap 95% confidence intervals for core metrics. For multiple-choice tasks we enforce exact-match scoring and de-duplication; for open-ended evaluation we use the official judge prompts without chain-of-thought. All baselines share identical sampling, token, and frame budgets.

Image benchmarks. Visual Jigsaw yields consistent improvements across fine-grained perception and spatial understanding. Table 5 summarizes results, including MMVP, MMStar, HR-Bench, and DA-2K. We also include comparisons to reasoning-oriented baselines.

Video benchmarks. We evaluate at 16/32/64-frame settings across diverse tasks (AoTBench, Vinoground, VideoMME, TempCompass, CVBench). Visual Jigsaw improves over the base model at all frame counts with stronger gains at 32–64 frames; see Table 6.

3D benchmarks. Visual Jigsaw improves depth comparison (DA-2K) and broader spatial reasoning (OmniSpatial, 3DSRBench, All-Angles, ViewSpatial-Bench). Largest gains appear on tasks that directly probe depth and near/far ordering; see Table 3.

Reasoning preservation. On a reasoning-oriented model (ThinkLite-VL (Wang et al., 2025b)), adding Visual Jigsaw for images preserves or slightly improves math and general reasoning while boosting perception (Table 4). This supports the claim that Visual Jigsaw complements reasoning-centric RL.

Table 1: **Evaluation results on image benchmarks.** Image Jigsaw achieves consistent improvements across fine-grained perception, spatial understanding, and compositional understanding tasks.

	Fine-grained Perception & Understanding								Spatial Und (Mono)			Compositional Und	
Model	**MMVP**	**MMStar (fine-grained)**	**MMBench**	**HR-Bench-8K**	**V***	**MME-RealWorld**	**LISA-Grounding**	**OVD-Eval**	**VSR**	**OmniSpatial**	**DA-2K**	**Winoground**	**SugarCrepe++**
	test	fine	en_dev	test	test	lite	test	test	test	test	val	g-acc	test
ThinkLite-VL	55.33	59.95	84.19	68.12	76.96	46.17	73.70	35.78	78.09	42.60	58.46	35.25	61.49
VL-Cogito	55.33	56.64	82.98	69.62	79.58	47.63	72.26	35.78	79.82	44.29	56.43	38.25	63.59
LLaVA-Critic-R1	53.33	57.80	83.16	67.50	78.01	45.18	68.52	35.28	78.50	42.73	53.82	34.75	61.93
Qwen2.5-VL-7B	54.66	59.75	83.33	67.38	76.96	43.41	71.89	35.07	77.68	42.66	54.45	37.00	61.59
Image Jigsaw (SFT)	56.00	60.94	83.67	69.75	80.10	43.88	66.59	34.35	80.68	43.55	61.46	38.75	62.03
Image Jigsaw	60.66	65.81	84.45	71.13	80.63	45.96	74.54	36.49	80.36	44.49	60.35	39.00	63.02
(Gain)	+6.00	+6.06	+1.12	+3.75	+3.66	+2.55	+2.65	+1.42	+2.68	+1.83	+5.90	+2.00	+1.43

Table 2: **Evaluation results on video benchmarks.** Video Jigsaw consistently improves over the baseline across all benchmarks and frame settings.

Model	Frames	**AoTBench**	**Vinoground**	**TOMATO**	**FAVOR-Bench**	**TUNA-Bench**	**VideoMME**	**TempCompass**	**TVBench**	**MotionBench**	**LVBench**	**VSI-Bench**	**Video-TT**	**CVBench**
		vqa	group	test	test	test	wo subs	mc	test	val	test	test	mcq	test
Video-R1	16	45.06	9.40	27.29	49.47	53.00	56.62	70.19	51.80	55.82	34.53	34.34	42.95	47.50
Video-R1	32	47.53	10.20	27.29	49.90	54.26	59.88	71.77	53.54	56.12	38.61	35.11	42.63	48.10
Video-R1	64	48.68	10.60	27.36	50.51	54.33	60.85	72.59	53.43	56.09	38.80	36.61	42.74	48.69
Qwen2.5-VL-7B	16	45.52	12.60	25.87	48.54	53.14	57.44	71.77	49.94	55.56	33.51	32.79	38.39	47.70
Qwen2.5-VL-7B	32	49.48	18.20	26.34	49.34	54.88	60.70	72.59	51.96	56.47	39.19	35.34	41.57	49.60
Qwen2.5-VL-7B	64	52.41	21.80	26.35	50.86	55.79	63.44	72.84	53.74	56.29	40.35	37.74	42.25	51.50
Video Jigsaw	16	51.67	15.20	27.56	49.69	55.10	58.07	73.10	51.33	56.87	36.41	35.39	40.19	49.80
(Gain)		+6.15	+2.60	+1.69	+1.15	+1.96	+0.63	+1.33	+1.39	+1.31	+2.90	+2.60	+1.80	+2.10
Video Jigsaw	32	55.00	21.40	28.03	50.56	56.49	62.37	73.60	53.31	57.99	39.70	38.47	43.27	51.60
(Gain)		+5.52	+3.20	+1.69	+1.22	+1.61	+1.67	+1.01	+1.35	+1.52	+0.51	+3.13	+1.70	+2.00
Video Jigsaw	64	57.64	25.20	28.30	52.27	56.63	64.74	73.60	54.18	57.91	41.83	40.40	44.11	54.50
(Gain)		+5.23	+3.40	+1.95	+1.41	+0.84	+1.30	+0.76	+0.44	+1.62	+1.48	+2.66	+1.86	+3.00

5 ANALYSIS

Why does Visual Jigsaw help? The jigsaw objective enforces sensitivity to local appearance, temporal continuity, and depth/geometry cues. Because rewards are verifiable and shaped, the policy receives dense, low-noise signals that emphasize perceptual correctness over linguistic priors.

Information-theoretic view. Permutation recovery pressures representations to encode edges, part boundaries, and motion continuity that predict order yet remain independent of captions, reducing language priors and aiding transfer to counting, compositionality, and relative depth while staying compatible with instruction tuning.

Ablations on jigsaw difficulty. Figure 4 studies patch/grid sizes and clip counts. We find monotonically increasing performance up to a moderate difficulty, after which optimization becomes less stable, illustrating a bias–variance trade-off in reward shaping. The graded reward (gamma 0.2) is critical to stabilize training at higher difficulty.

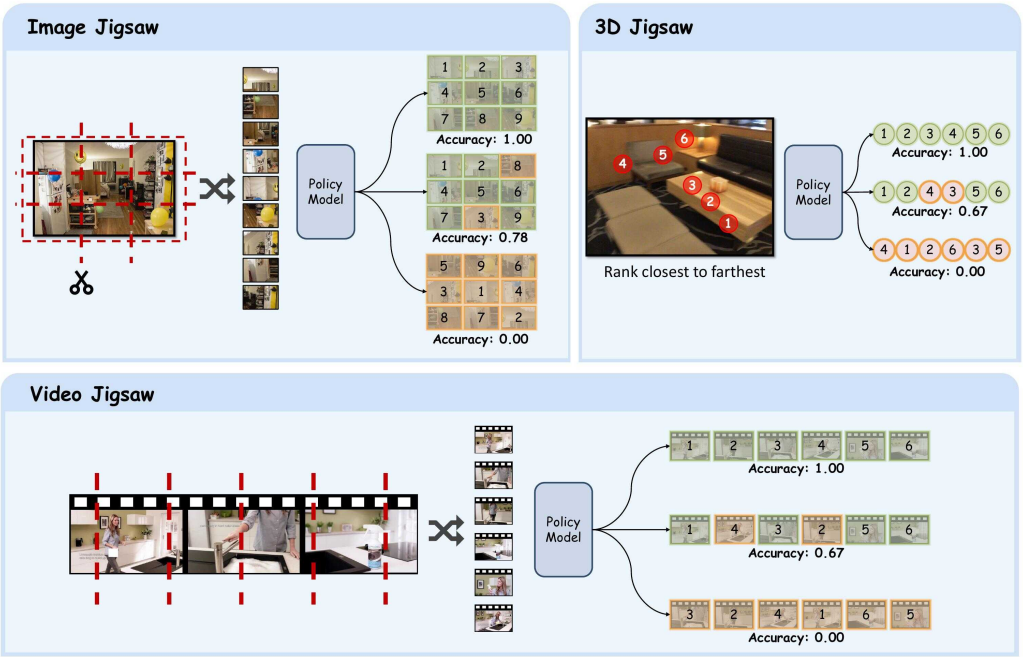

Figure 3: Method overview and verifiable reward.

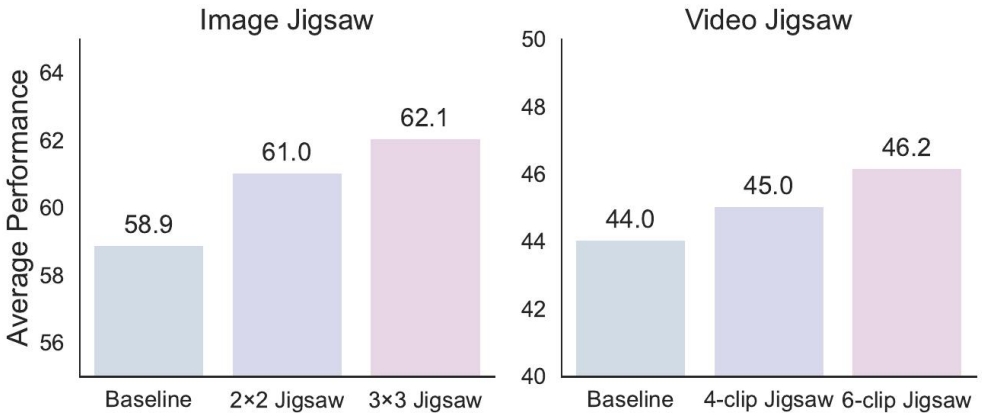

Figure 4: Performance with different jigsaw difficulties on image and video tasks.

Table 3: **Evaluation results on 3D benchmarks.** 3D Jigsaw consistently enhances performance on both directly related depth comparison tasks (DA-2K) and broader 3D perception tasks spanning single-view, multi-view, and egocentric video inputs.

Model	SAT-Real	3DSRBench	ViewSpatial	All-Angles	OmniSpatial	VSI-Bench	SPARBench	DA-2K
	test	test	test	test	test	test	tiny	test
Qwen2.5-VL-7B	48.66	57.42	36.52	47.56	42.66	37.74	35.75	54.45
3D Jigsaw	64.00	58.13	38.62	49.06	45.99	40.64	38.31	71.56
(Gain)	+15.34	+0.71	+2.10	+1.50	+3.33	+2.90	+2.56	+17.11

Table 4: Performance of Visual Jigsaw on a reasoning-oriented MLLM (ThinkLite-VL), showing improved visual perception while preserving reasoning ability.

Model	Vision-Centric Avg	MathVista testmini	MathVision testmini	MathVerse testmini	MMMU val	EMMA mini
ThinkLite-VL	59.69	75.20	30.92	50.76	55.11	26.75
+ Image Jigsaw	61.60	75.10	35.20	50.50	54.22	29.00

Reward shaping. The discounted agreement term provides usable gradients when predictions are close but not exact, reducing brittleness to local swaps and accelerating convergence at larger K. We observe lower variance and more stable updates under this shaping, consistent with the curriculum results in Figure 4.

RL vs SFT. RL consistently outperforms SFT on perception tasks; in our image experiments, SFT improves some metrics but degrades others (e.g., LISA-Grounding 66.59 with SFT vs 74.54 with RL in Table 5). We hypothesize SFT overfits to surface patterns of permutation formats, whereas RL with verifiable rewards better internalizes perceptual constraints.

Cross-modal transfer. Training on one modality improves related abilities in others (e.g., image jigsaw improves compositionality, video jigsaw improves temporal order reasoning). The effect is most pronounced when the target benchmark aligns with the jigsaw's core signal (e.g., depth ordering for DA-2K).

Error analysis. Qualitative behavior. Successful policies group semantically coherent parts, respect motion continuity, and reassemble occluded objects; failures occur on symmetric layouts and low-texture regions where multiple orders fit the evidence. Failures cluster on symmetric layouts, near-duplicate clips, and reflective or textureless regions. Format penalties prune spurious strings, while partial credit guides locally consistent permutations.

Design choices and invariances. Two properties appear central to Visual Jigsaw's effectiveness: (i) permutation supervision focuses the model on relative spatial and temporal relations rather than absolute coordinates, and (ii) the task admits exact verification from the input without extra annotations. The first encourages locality-aware features (edges, part boundaries, motion continuity) that transfer to downstream perception benchmarks; the second yields low-noise rewards that align optimization pressure with perceptual correctness. In practice we randomize global crops and flips, normalize patch sizes across resolutions, and canonicalize the textual permutation to a single delimiter and ordering. These choices reduce spurious shortcuts, improve reproducibility across seeds, and make the policy less sensitive to prompt phrasing, suggesting that the gains stem from the verification signal rather than prompt engineering.

Sample efficiency and compute. Visual Jigsaw keeps sequence length unchanged; reward scoring adds a small constant overhead (¡5% step time). In our runs, mid-scale models reached stable gains within 50k–100k steps; larger corpora improved more slowly and saturated at similar difficulty.

Table 5: **Evaluation results on image benchmarks.** Image Jigsaw achieves consistent improvements across fine-grained perception, spatial understanding, and compositional understanding tasks.

	Fine-grained Perception & Understanding								Spatial Und (Mono)			Compositional Und	
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Qwen2.5-VL-7B	54.66	59.75	83.33	67.38	76.96	43.41	71.89	35.07	77.68	42.66	54.45	37.00	61.59
Image Jigsaw (SFT)	56.00	60.94	83.67	69.75	80.10	43.88	66.59	34.35	80.68	43.55	61.46	38.75	62.03
Image Jigsaw	60.66	65.81	84.45	71.13	80.63	45.96	74.54	36.49	80.36	44.49	60.35	39.00	63.02
(Gain)	+6.00	+6.06	+1.12	+3.75	+3.66	+2.55	+2.65	+1.42	+2.68	+1.83	+5.90	+2.00	+1.43

Table 6: **Evaluation results on video benchmarks.** Video Jigsaw consistently improves over the baseline across all benchmarks and frame settings.

Model	Frames	**AoTBench**	**Vinoground**	**TOMATO**	**FAVOR-Bench**	**TUNA-Bench**	**VideoMME**	**TempCompass**	**TVBench**	**MotionBench**	**LVBench**	**VSI-Bench**	**Video-TT**	**CVBench**
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Video Jigsaw	64	57.64	25.20	28.30	52.27	56.63	64.74	73.60	54.18	57.91	41.83	40.40	44.11	54.50
(Gain)		+5.23	+3.40	+1.95	+1.41	+0.84	+1.30	+0.76	+0.44	+1.62	+1.48	+2.66	+1.86	+3.00

Per-category trends. Largest gains appear on fine-grained localization, counting, and relative depth; OCR-heavy or long-horizon reasoning changes are smaller. Improvements scale with exposure to local order cues (edges, motion), echoing Figure 4.

Failure cases and mitigations. Symmetries and repeating textures can induce globally flipped permutations. Mixing patch aspect ratios and penalizing equivalent-format strings reduces such errors, though low-light videos remain challenging.

Curriculum and stability. We stage difficulty by increasing K only after the policy reaches a target exact match rate on the current level, keeping the same prompt template. A brief format-only phase trains the parser and output discipline before enabling partial credit. Group-normalized updates and a small entropy bonus discourage degenerate permutations, and we cap per-prompt retries to prevent reward hacking via sampling. These choices reduce variance and make training reproducible across seeds. Ranking perspective. The jigsaw task can be viewed as learning to predict a permutation that minimizes a rank-based discrepancy with the ground truth. The discounted agreement term acts like a top-heavy ranking utility: early positions contribute more than later ones, encouraging correct

anchors that propagate global order. This explains why partial rewards improve sample efficiency on large K while preserving a unique optimum at exact match.

6 Conclusion

We present Visual Jigsaw, a simple, scalable, and modality-general post-training task that strengthens intrinsic visual understanding in MLLMs without architectural changes or generative supervision. By casting perception as permutation recovery with verifiable rewards and graded shaping, the method improves fine-grained image perception, video temporal understanding, and 3D spatial reasoning while preserving reasoning. Results across images, videos, and 3D show consistent transfer on diverse benchmarks, and comparisons to SFT and reasoning-centric RL indicate complementary benefits. We hope Visual Jigsaw encourages broader exploration of self-supervised, vision-centric objectives for aligning MLLMs.

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